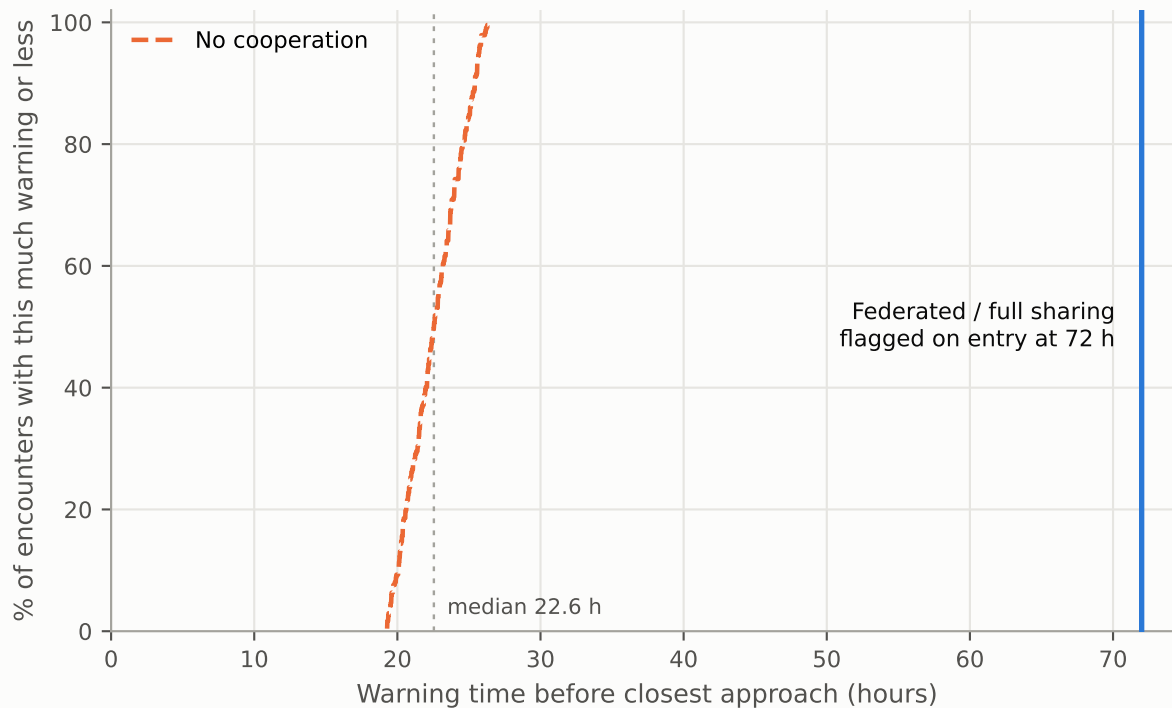


How late no-cooperation is, across 237 real threshold-crossing encounters



10,713 real Starlink objects over 168h, screened 2026-09-20T08:46:01+00:00. Orbits, geometry and Pc unmodified; ownership assigned by NORAD parity. Cooperation buys a median 49.5 extra hours of warning — federated at 0% raw-data exposure, full sharing at 100%.